gathering, especially during the Cold War and Vietnam War. UAVs such as the *Ryan Model 147* "*Lightning Bug*" and *IAI Scout* provided real-time imagery and electronic intelligence (ELINT), allowing armed forces to monitor enemy movements and assess damage after strikes without risking pilots.

- *Target practice and gunnery training*: UAVs like the *Ryan Firebee* were used as aerial targets for missile testing and anti-aircraft gunnery exercises, offering reusable and cost-effective platforms for training.
- *Electronic warfare (EW) and communication relay*: Drones equipped with electronic countermeasure payloads and signal relay systems supported communication networks and disrupted enemy radar or missile systems during combat operations.
- *Border patrol and coastal surveillance*: UAVs such as the *RQ-2 Pioneer* were employed by the US Navy and Marine Corps for maritime reconnaissance and coastal defense, using real-time video feeds to detect and track vessels.

3.4.2 Scientific and environmental applications

- *Meteorological and atmospheric studies*: Equipped with sensors for temperature, pressure, and humidity measurement, UAVs were used in weather monitoring, storm tracking, and atmospheric sampling, providing valuable data for climate research.
- *Environmental and wildlife monitoring*: Drones facilitated aerial observation of wildlife habitats, deforestation, and environmental degradation, supporting conservation and ecological research efforts.
- *Disaster management and emergency response*: UAVs were employed in post-disaster assessments for mapping affected areas, locating